

Highlights

Active Contact Localization Without Joint-Torque or Force/Torque Sensing: Ambiguity-Preserving Inference for Robot Manipulators

- Contact location and force are measured from joint states and commanded torques only.
- A single view determines a line; two transverse views provably determine the point.
- Closed-form information bound links accuracy to force magnitude and view separation.
- Line-of-action support preservation suppresses confident-but-wrong estimates.
- A GUM-type residual uncertainty budget quantifies the sensorless measurement.

Active Contact Localization Without Joint-Torque or Force/Torque Sensing: Ambiguity-Preserving Inference for Robot Manipulators

ARTICLE INFO

Keywords:

Sensorless contact localization
Momentum observer
Observability
Particle filter
Active measurement
Measurement uncertainty

ABSTRACT

We measure the location of an external contact on an unknown grasped object, together with the contact force, from the joint positions, velocities, and commanded torques of a 7-DoF manipulator, without joint-torque, force/torque, or tactile sensors or an object model. The difficulty is structural rather than stochastic: from a single arm configuration, every point on the force line of action produces the same joint residual. We prove that two orientations with transverse object-frame force directions identify the contact point uniquely, and derive a closed-form information bound that scales with the force magnitude and the sine of half the interview force angle. The proposed method validates the payload-compensated momentum-observer residual with a frozen trust gate, expresses its uncertainty with a GUM-type budget that includes follower-force variation, represents the ambiguity with a force-profiled particle filter, preserves hypotheses along the unresolved line of action, and selects the next orientation by profile Fisher information inside the validated envelope. In 1,200 paired trials on a simulated Franka Research 3 with realistic model and actuation errors, ambiguity-preserving recovery reduces the 95th-percentile localization error from 137.6 mm to 21.9 mm and the confident-but-wrong rate from 17.7% to 0.8%, with median position and force-magnitude errors of 6.6 mm and 1.3%; after two views, the actively selected orientation reduces the 95th-percentile error from 40.4 to 25.9 mm relative to a random one. **On the real robot, with an independently measured pusher, the method attains a median position error of 8.7 mm over 96 trials.** Project page: <https://anonymous.4open.science/w/apcl/>.

1. Introduction

A manipulator that knows where and how hard it is being touched can react safely to people, comply with a guiding hand, and reason about the objects it holds. Measuring this contact information normally requires dedicated instrumentation: joint-torque sensors [1], a wrist force/torque (F/T) sensor [2], or tactile skins. Many industrial and low-cost arms carry none of these, and a grasped tool or object extends the contact surface beyond any skin that the arm might carry. This paper treats contact localization as an indirect measurement problem: the joint encoders and the commanded torque already available in every torque-controlled arm constitute the measurement chain, and the task is to extract the contact point on an unknown grasped object and the contact force from them, together with an uncertainty statement that can be trusted [3].

Momentum observers recover the external joint torque from proprioceptive signals without acceleration measurement [1, 4, 5], data-driven detectors flag collision events from internal robot signals [6], and contact particle filters turn observer residuals into contact-location posteriors [7, 8]. The key obstacle to localizing a contact on an unknown object, however, is not observer noise. A single contact force $\mathbf{f}$ applied at point $\mathbf{p}$ produces the moment $\mathbf{p} \times \mathbf{f}$, and every point $\mathbf{p} + \alpha \mathbf{f}$ produces exactly the same moment. From a single arm configuration, the joint residual therefore constrains the contact to a line, whatever the number of joints, the observer bandwidth, or the sensor resolution. This line-of-action ambiguity is classical in intrinsic contact sensing [2, 9], where it is removed by assuming a known surface. With an unknown object, no surface is available to intersect the line.

We remove the ambiguity by changing the measurement configuration instead of adding a sensor. Rotating the grasped object about its current position changes the object-frame force direction while the same material point stays in contact, so a second view yields a second line that intersects the first at the true contact. Exploiting this idea with a finite-sample estimator raises a reliability issue that we identify as the dominant failure mode. Between views, the posterior is legitimately line-shaped, but the particles that lie on the line are few, and resampling concentrates them on a random stretch of it. The particles near the true contact may disappear before the second view arrives, and the filter then converges to a narrow posterior that excludes the truth: a *confident-but-wrong* (CBW) measurement. A narrow posterior is useful only if it contains the true contact.

Our approach is summarized as *preserve first, disambiguate actively*. The contributions are:

1. **Measurement observability.** We prove that the single-view contact measurement is non-identifiable exactly along the object-frame force direction, that two views identify the contact point and the world-frame force uniquely if and only if their object-frame force directions are not parallel, and we give a closed-form Fisher-information bound in terms of force magnitude and inter-view angle (Section 3.2).
2. **Calibrated sensorless residual.** We combine a compact payload calibration and a frozen trust gate with a GUM-type uncertainty budget whose Type A part is evaluated from held-out calibration residuals and whose Type B part covers friction, torque-constant tolerance, and follower-force variation (Section 3.3).
3. **Ambiguity-preserving inference.** We introduce a force-profiled contact particle filter whose recovery layer distinguishes weight degeneracy, model inconsistency, and loss of support along the unresolved line of action, and re-supplies hypotheses along the theoretical line of action whenever the accumulated information along it is insufficient (Section 3.5).
4. **Active measurement design.** We select the next orientation inside the validated envelope by maximizing the posterior-expected profile Fisher information of the contact point, with the force as a nuisance parameter, under a position-preserving motion that keeps the follower contact intact (Section 3.6).

We quantify accuracy, tail error, CBW rate, and credible-region coverage in 1,200 paired simulation trials, and show that recovery is what makes the recursive measurement reliable and active sensing what makes it efficient (Section 4). A physical-contact study then establishes what the pusher must provide for the measurement to hold: tangential compliance (Section 4.7).

2. Related work

External torque estimation Generalized-momentum observers [4, 5] estimate the external joint torque as a first-order filtered residual and underpin collision detection, isolation, and identification [1]. Cartesian contact wrenches have been estimated from motor currents [10], from the observer residual under an assumed contact location [11, 12], and from identified dynamics on hydraulic manipulators [13]. Unmodeled dynamics and actuator faults contaminate such residuals and can be learned with dedicated identification [14]; friction in particular is a dominant residual component [15]. These methods measure the wrench when the contact location is known; they do not locate the contact on an unknown object.

Contact localization Intrinsic contact sensing recovers the contact point from a wrist F/T measurement and a known surface [2, 9]. On the robot body, the contact link and point can be isolated from the residual [1, 16], sensing redundancy improves localization on torque-sensing robots [17], and the contact particle filter [7] localizes contacts on a known robot mesh from proprioception. Fed with the generalized-momentum residual computed from motor currents and encoders, the contact particle filter also localizes contacts on a collaborative robot without joint-torque sensors [8]. Learning-based localization maps proprioceptive signals to contact points [18], including on arms in motion without joint-torque measurements [19]. All of these rely on known geometry, which removes the line-of-action ambiguity by intersection. Even then, distinct contacts on the surface can produce nearly identical joint torques, so the attainable estimate is a set of candidate points, whose spurious members are falsified by small exploratory motions [20]. We address an unknown grasped object, where the ambiguity must be represented and resolved.

Payload identification The inertial parameters of a load are linear in the joint-space dynamics [21] and can be identified online for unknown held objects [22], or directly with wrist or fingertip F/T sensors [23]. Regularization and physical-consistency constraints keep the identified parameters feasible under limited excitation [24, 25]. We need only mass and center of mass and identify them from a compact, contact-free orientation sweep.

Particle degeneracy Sample impoverishment in particle filters is usually monitored by the effective sample size and countered by resampling and kernel roughening [26, 27]. These criteria detect a collapse of the weights, but not a loss of support along a direction that the current measurement cannot resolve; our results show that the latter dominates the tail error in contact localization.

Active sensing and experiment design Information-based selection of measurement configurations is standard in optimal experiment design [28] and in robot identification [29], and interactive perception exploits deliberate motion to reveal hidden properties [30]. Tactile localization with particle filters has used information gain to choose touches

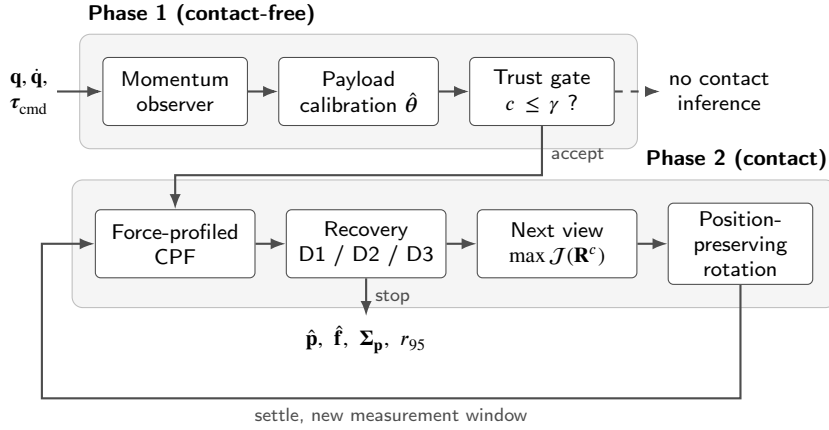


Figure 1: Sensorless contact measurement pipeline. Phase 1 is contact-free; Phase 2 alternates measurement windows, inference, and reorientation.

Table 1

Measurement availability.

Signal	Used by estimator	Role
Joint position q , velocity $\dot{q}$	Yes	Input
Commanded joint torque τ_{cmd}	Yes	Input
Measured joint torque	No	—
Wrist F/T sensor	No	—
Tactile sensing	No	—
Object geometry	No	—
Contact point and force	No	Ground truth only

[31]. In visually occluded scenes, contact-rich rummaging plans robot trajectories online by maximizing their mutual information with a particle belief over the object pose [32]. Unlike exploratory motions that falsify a finite set of candidate contacts on a known surface [20], our criterion scores each reorientation against the continuous line-of-action posterior of an unknown object. We apply the same principle to proprioceptive contact localization, with a profile information measure that eliminates the unknown force.

3. Sensorless contact measurement

This section defines the measurement model, establishes what it can and cannot identify, and derives the measurement method from this analysis. Fig. 1 summarizes the method. Phase 1 runs before contact and delivers a payload-compensated residual that has passed a model-validity test, together with its uncertainty budget (Section 3.3). Phase 2 runs after contact onset and alternates quasi-static measurement windows, particle updates with recovery, and information-driven reorientation, without any reset of the robot or observer state between phases (Sections 3.4–3.6).

3.1. Measurement model

A fixed-base 7-DoF manipulator holds an unknown object rigidly. Let $\{E\}$ be a frame attached to the end effector and the object, and let $R_k \in SO(3)$ and $o_k \in \mathbb{R}^3$ be its orientation and position in the world frame $\{W\}$ at measurement pose k . The available measurements and the unavailable ones are listed in Table 1. The unknowns are the contact point $p \in \mathcal{B} \subset \mathbb{R}^3$ expressed in $\{E\}$, where $\mathcal{B}$ is a bounded search region around the end effector, and the world-frame contact force $f \in \mathbb{R}^3$.

We consider the following contact regime. (A1) A single contact acts at one material point p of the object throughout an episode. (A2) The world-frame force f is constant over the episode up to a small follower variation, which is modeled explicitly in Section 3.3. (A3) The contact is a follower contact, such as a human hand or a force-controlled pusher that is compliant tangentially to the surface, which stays on the same material point while the object is reoriented. (A4) Measurements used for localization are taken in quasi-static windows after each reorientation has settled.

Residual model The generalized-momentum observer [4, 5]

$$\mathbf{r}(t) = \mathbf{K}_O \left(\mathbf{M}(\mathbf{q})\dot{\mathbf{q}} - \int_0^t (\boldsymbol{\tau}_{\text{cmd}} + \mathbf{C}^\top(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{g}(\mathbf{q}) - \mathbf{D}\dot{\mathbf{q}} + \mathbf{r}) \, ds \right), \quad (1)$$

where $\mathbf{D}$ is the nominal viscous joint damping, satisfies $\dot{\mathbf{r}} = \mathbf{K}_O(\boldsymbol{\tau}_{\text{dist}} - \mathbf{r})$, i.e., $\mathbf{r}$ is a first-order low-pass estimate of the total disturbance torque. Evaluated with the nominal model of the arm, this disturbance contains the external contact torque, the gravity torque of the unknown payload, model and actuation errors, and noise:

$$\mathbf{r} = \boldsymbol{\tau}_{\text{ext}} + \mathbf{Y}(\mathbf{q})\boldsymbol{\theta} + \boldsymbol{\delta} + \boldsymbol{\epsilon}. \quad (2)$$

The payload term is compensated in Phase 1, and the error terms are screened by the trust gate and quantified by the uncertainty budget (Section 3.3).

Let $\mathbf{J}_k \in \mathbb{R}^{6 \times 7}$ be the body Jacobian at pose k that maps joint velocities to the twist of $\{E\}$ expressed in $\{E\}$. The contact force expressed in $\{E\}$ is $\mathbf{f}_k = \mathbf{R}_k^\top \mathbf{f}$, and the equivalent wrench at the origin of $\{E\}$ is $\mathbf{w}_k = [\mathbf{f}_k^\top, (\mathbf{p} \times \mathbf{f}_k)^\top]^\top$. Hence

$$\boldsymbol{\tau}_{\text{ext},k} = \mathbf{J}_k^\top \mathbf{w}_k = \underbrace{\mathbf{J}_k^\top \begin{bmatrix} \mathbf{I}_3 \\ [\mathbf{p}]_\times \end{bmatrix} \mathbf{R}_k^\top}_{\mathbf{G}_k(\mathbf{p})} \mathbf{f}, \quad (3)$$

where $[\cdot]_\times$ is the skew-symmetric matrix of the cross product. After payload compensation, the measurement model is $\mathbf{r}_k = \mathbf{G}_k(\mathbf{p})\mathbf{f} + \boldsymbol{\epsilon}_k$. It is linear in the force and bilinear in the contact point and force.

3.2. Observability

Proposition 1 (Single-view line-of-action ambiguity). *For any pose k , any $(\mathbf{p}, \mathbf{f})$, and any $\alpha \in \mathbb{R}$, $\mathbf{G}_k(\mathbf{p} + \alpha \mathbf{f}_k) \mathbf{f} = \mathbf{G}_k(\mathbf{p}) \mathbf{f}$. Consequently, the single-view likelihood is constant along the line $\{\mathbf{p} + \alpha \mathbf{f}_k\}$, and no estimator can resolve the contact point along $\mathbf{f}_k$ from pose k alone.*

PROOF. $[\mathbf{p} + \alpha \mathbf{f}_k]_\times \mathbf{f}_k = \mathbf{p} \times \mathbf{f}_k + \alpha \mathbf{f}_k \times \mathbf{f}_k = \mathbf{p} \times \mathbf{f}_k$.

Proposition 1 holds for any number of joints and any noise level. It carries over to the profile likelihood in which the force is estimated: if $\hat{\mathbf{p}}$ maximizes the single-view profile likelihood and $\hat{\mathbf{f}}$ is its profiled force, every point $\hat{\mathbf{p}} + \alpha \hat{\mathbf{f}}_k$ attains the same maximal value, because $\hat{\mathbf{f}}$ reproduces the same fitted residual there. The single-view profile likelihood thus has a flat ridge along a whole line instead of a peak. A line-shaped single-view posterior is therefore not an estimator failure; it is the correct representation of the single-view inverse problem.

Proposition 2 (Two-view identifiability). *Let $\mathbf{J}_1$ and $\mathbf{J}_2$ have full row rank. The noise-free measurements $\mathbf{G}_1(\mathbf{p})\mathbf{f}$ and $\mathbf{G}_2(\mathbf{p})\mathbf{f}$ determine $(\mathbf{p}, \mathbf{f})$ with $\mathbf{f} \neq \mathbf{0}$ uniquely if and only if $\mathbf{f}_1 \times \mathbf{f}_2 \neq \mathbf{0}$. Let the relative rotation $\Delta \mathbf{R} = \mathbf{R}_2 \mathbf{R}_1^\top$ be a rotation by the angle $\varphi \in (0, \pi]$ about the unit axis $\mathbf{a}$. Then the condition holds if and only if $\mathbf{a}$ is not parallel to $\mathbf{f}$ and $\Delta \mathbf{R}$ is not a half-turn about an axis perpendicular to $\mathbf{f}$.*

PROOF. Full row rank of $\mathbf{J}_k$ makes $\mathbf{J}_k^\top$ injective, so each view determines the wrench $\mathbf{w}_k$. Suppose $(\mathbf{p}, \mathbf{f})$ and $(\mathbf{p}', \mathbf{f}')$ give identical wrenches. The force components give $\mathbf{R}_k^\top \mathbf{f} = \mathbf{R}_k^\top \mathbf{f}'$, hence $\mathbf{f} = \mathbf{f}'$. The moment components give $(\mathbf{p} - \mathbf{p}') \times \mathbf{f}_k = \mathbf{0}$ for $k = 1, 2$, hence $\mathbf{p} - \mathbf{p}' \in \text{span}(\mathbf{f}_1) \cap \text{span}(\mathbf{f}_2)$, which is $\{\mathbf{0}\}$ if and only if $\mathbf{f}_1 \times \mathbf{f}_2 \neq \mathbf{0}$. If $\mathbf{f}_1 \parallel \mathbf{f}_2$, then $(\mathbf{p} + \alpha \mathbf{f}_1, \mathbf{f})$ reproduces both measurements for every α . Finally, $\mathbf{f}_1 \parallel \mathbf{f}_2$ if and only if $\Delta \mathbf{R}^\top \mathbf{f} \parallel \mathbf{f}$, i.e., $\mathbf{f}$ is an eigenvector of $\Delta \mathbf{R}$, which occurs exactly when $\mathbf{a} \parallel \mathbf{f}$ or when $\varphi = \pi$ with $\mathbf{a} \perp \mathbf{f}$.

Remark 1 (Per-view forces). *The proof uses the force components only to fix the object-frame force of each view. Proposition 2 therefore also holds, with the condition $\mathbf{f}_1 \times \mathbf{f}_2 \neq \mathbf{0}$, when every view has its own unknown force $\mathbf{f}_k$ and only the contact point is shared: each view determines its own line of action, and two non-parallel lines intersect at the contact. Assumption (A2) is thus not what makes the contact point identifiable; it lets the estimator pool the force information of all views, which tightens the estimate.*

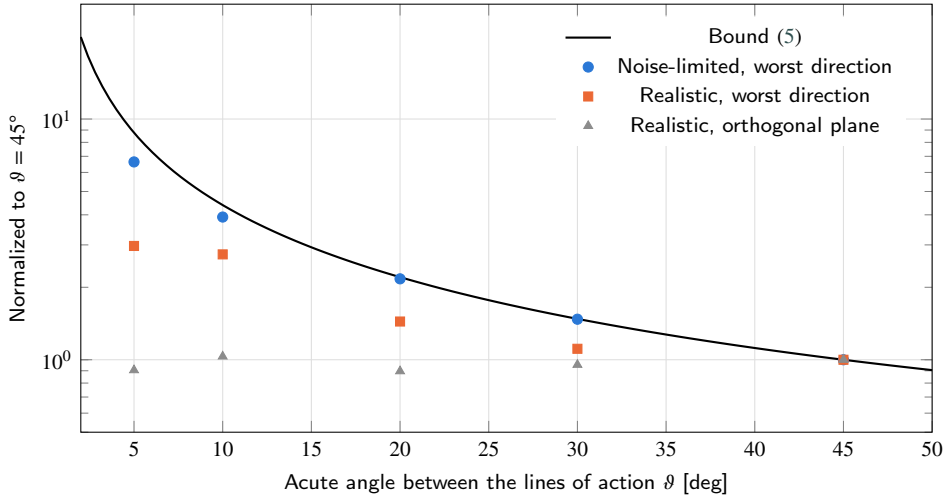


Figure 2: Two-view Cramér–Rao bound on the worst-direction contact-point standard deviation versus the acute angle between the lines of action, and the simulated RMS two-view errors along the worst direction and in the orthogonal plane, all normalized to $\theta = 45^\circ$. The noise-limited study follows the bound; under realistic errors, the worst-direction error still rises as the lines approach each other, while the orthogonal error stays at the bias floor.

Rotating the object about the force line therefore conveys no information, whereas any rotation with a component perpendicular to the force does. The payload calibration of Section 3.3 is the same problem with gravity as a known-direction force acting at the unknown center of mass, and Proposition 2 explains why it requires orientation diversity about horizontal axes.

The following bound quantifies how the accuracy grows with view separation. Consider the moment channel with isotropic noise $\sigma_m^2 \mathbf{I}_3$ and the force treated as known. The Fisher information on $\mathbf{p}$ from K views is

$$\mathbf{I}_p = \frac{1}{\sigma_m^2} \sum_{k=1}^K [\mathbf{f}_k]_{\times}^T [\mathbf{f}_k]_{\times} = \frac{1}{\sigma_m^2} \sum_{k=1}^K (\|\mathbf{f}\|^2 \mathbf{I}_3 - \mathbf{f}_k \mathbf{f}_k^T). \quad (4)$$

Corollary 1 (Two-view information bound). Let $K = 2$, and let $\vartheta \in [0, \pi/2]$ be the acute angle between the two lines of action, $\cos \vartheta = |\hat{\mathbf{f}}_1^T \hat{\mathbf{f}}_2|$. The eigenvalues of $\mathbf{I}_p$ are $\|\mathbf{f}\|^2(1 \pm \cos \vartheta)/\sigma_m^2$ and $2\|\mathbf{f}\|^2/\sigma_m^2$, and the worst-direction Cramér–Rao bound on the contact-point standard deviation is

$$\sigma_p^{\text{worst}} \geq \frac{\sigma_m}{\sqrt{2} \|\mathbf{f}\| \sin(\vartheta/2)}, \quad \cos \vartheta = \left| \cos \varphi + (1 - \cos \varphi)(\mathbf{a}^T \hat{\mathbf{f}})^2 \right|. \quad (5)$$

PROOF. The matrix $\hat{\mathbf{f}}_1 \hat{\mathbf{f}}_1^T + \hat{\mathbf{f}}_2 \hat{\mathbf{f}}_2^T$ has eigenvalues $1 \pm \hat{\mathbf{f}}_1^T \hat{\mathbf{f}}_2$ and 0. Substituting into (4) gives the eigenvalues $\|\mathbf{f}\|^2(1 \mp \hat{\mathbf{f}}_1^T \hat{\mathbf{f}}_2)/\sigma_m^2$ and $2\|\mathbf{f}\|^2/\sigma_m^2$, whose minimum is $\|\mathbf{f}\|^2(1 - \cos \vartheta)/\sigma_m^2 = 2\|\mathbf{f}\|^2 \sin^2(\vartheta/2)/\sigma_m^2$ for the acute angle ϑ . The angle relation follows from $\mathbf{f}_1^T \mathbf{f}_2 = \mathbf{f}^T \Delta \mathbf{R}^T \mathbf{f}$ and Rodrigues' formula.

Because a line of action has no direction, the bound depends only on the acute angle between the two lines: it is inversely proportional to the force magnitude and to $\sin(\vartheta/2)$, and for a rotation about an axis perpendicular to the force, ϑ equals the rotation angle φ up to 90° and $\pi - \varphi$ beyond. In the estimator, the force is unknown and acts as a nuisance parameter; Section 3.6 uses the corresponding profile information, which accounts for the force uncertainty in the same way that precision bounds account for unknown systematic effects [33]. Fig. 2 compares the bound with the simulated two-view errors of Section 4.

3.3. Validated residual and uncertainty budget

Payload calibration Under (A4), inertial effects of the payload are negligible in the measurement windows, and its contribution to (2) is the gravity torque

$$\mathbf{Y}(q)\boldsymbol{\theta} = \mathbf{J}^T \begin{bmatrix} \mathbf{R}^T \mathbf{g}_W & \mathbf{0} \\ \mathbf{0} & -[\mathbf{R}^T \mathbf{g}_W]_{\times} \end{bmatrix} \begin{bmatrix} m \\ m\mathbf{c} \end{bmatrix}, \quad \boldsymbol{\theta} = [m, m\mathbf{c}_x, m\mathbf{c}_y, m\mathbf{c}_z]^T, \quad (6)$$

Table 2Residual uncertainty budget for joint i , Eq. (8).

Component	Type	Evaluation
Sub-window noise s_i^2/n	A	Pooled variance of 0.1-s means in calibration windows
Model error $\max(\sigma_i^{\text{LOO}}, \sigma_i^{\text{ch}})$	A	Leave-one-out and challenge residual RMS
Residual dry friction $a_i/\sqrt{3}$	B	Uniform, $a_i = 0.06 / 0.03$ N m (joints 1–4 / 5–7)
Torque constant $u_\kappa \bar{r}_i$	B	Relative tolerance $u_\kappa = 1\%$
Follower-force variation $\varepsilon_f \ \mathbf{f}\ $	B	Per view, isotropic, $\varepsilon_f = 3\%$, Eq. (9)

which is linear in the four parameters [21, 22]. Restricting the identification to these four parameters is consistent with the quasi-static measurement protocol; the six inertia parameters are not excited by the measurement windows and are therefore neither needed nor identifiable from them. We estimate θ by ridge-regularized least squares from $N_c = 7$ contact-free orientations: the initial orientation and six tilts of 20° about the horizontal axes $\pm \mathbf{e}_x$, $\pm \mathbf{e}_y$, and $\pm(\mathbf{e}_x + \mathbf{e}_y)/\sqrt{2}$, with the end-effector position held fixed. The compact sweep avoids large workspace motions that would disturb the grasp, and by Proposition 2 its tilts about two horizontal directions make θ identifiable. Each orientation is reached in 1.5 s, allowed to settle for 0.5 s, and measured over a 1.0 s window.

Trust gate A small calibration residual shows that the regression interpolates the calibration poses, not that the model is valid: the four parameters can absorb a structured model error on a compact pose set. We therefore test the frozen calibration on $N_h = 3$ challenge orientations tilted by 35° about axes that are not used for fitting, with the statistic

$$c = \max_{j \in C} \frac{1}{n} \left\| \frac{\mathbf{r}_j - \mathbf{Y}_j \hat{\theta}}{\sigma_g} \right\|^2, \quad (7)$$

where $\sigma_g = 0.1$ N m is a fixed absolute scale and $n = 7$, and admit Phase 2 only if $c \leq \gamma$. The threshold γ is chosen once on a derivation set with offline validity labels, by the lexicographic rule: zero admitted CBW localizations, then zero admitted invalid models, then maximal acceptance of valid models. It is then frozen and evaluated on a disjoint held-out set. If the gate rejects, contact localization is not started. The challenge orientations also define the validated envelope within which Phase 2 is allowed to move.

Residual uncertainty budget A sensorless estimate is useful only if its stated uncertainty can be trusted [3, 34]. We express the standard uncertainty of the payload-compensated window-mean residual of joint i as a GUM-type budget,

$$u_i^2 = \underbrace{\frac{s_i^2}{n} + \max(\sigma_i^{\text{LOO}}, \sigma_i^{\text{ch}})^2}_{\text{Type A}} + \underbrace{\frac{a_i^2}{3} + (u_\kappa \bar{r}_i)^2}_{\text{Type B}}, \quad (8)$$

whose components are listed in Table 2. The Type A part is evaluated from Phase 1 data: s_i^2 is the pooled variance of the 0.1-s sub-window means within the calibration windows, n is the number of sub-windows averaged, and the model-error scale is the larger of the leave-one-out calibration residual σ_i^{LOO} and the challenge residual σ_i^{ch} , so that it reflects extrapolation to orientations not used for fitting. The Type B part covers effects that are invisible in contact-free calibration: residual dry friction after compensation, modeled as uniform with half-width a_i [15], and the tolerance u_κ of the torque constant, which scales the load-dependent part $\bar{r}_i$ of the residual.

The follower contact of (A3) does not reproduce the force exactly from one view to the next. We therefore model the force of view k as $\mathbf{f} + \delta_k$ with $\delta_k \sim \mathcal{N}(\mathbf{0}, \sigma_f^2 \mathbf{I}_3)$, $\sigma_f = \varepsilon_f \|\hat{\mathbf{f}}\|$, and marginalize δ_k in closed form in the likelihood of Section 3.4.

3.4. Force-profiled contact particle filter

Each particle $\mathbf{p}^{(i)} \in \mathcal{B}$, $i = 1, \dots, N$, represents a contact-point hypothesis. The force is not sampled but eliminated by generalized least squares, with a single world-frame force shared by all views. For view k with window-mean residual $\bar{\mathbf{r}}_k$, weight $\mathbf{W}_k = \text{diag}(u_i^{-2})$, and $\mathbf{A}_k = \mathbf{J}_k^\top$, define $\mathbf{T}_k(\mathbf{p}) = \mathbf{B}(\mathbf{p})^\top \mathbf{A}_k^\top \mathbf{W}_k \mathbf{A}_k \mathbf{B}(\mathbf{p})$ and $\mathbf{b}_k(\mathbf{p}) = \mathbf{B}(\mathbf{p})^\top \mathbf{A}_k^\top \mathbf{W}_k \bar{\mathbf{r}}_k$,

Table 3

Recovery rules. $\hat{N}_{\text{eff}} = 1 / \sum_i (w^{(i)})^2$; $\min_i S$ is the best profile residual; $\sigma_{\parallel}$ is the Cramér–Rao standard deviation along the current line of action.

Rule	Failure mode	Trigger	Re-supply
D1	Weight degeneracy	$\hat{N}_{\text{eff}} / N < \rho_{\text{th}}$, every update	Systematic resampling; fraction η jittered locally
D2	Narrow but inconsistent	$r_{95} < r_{\text{nar}}$ and $\min_i S > \chi_{0.99}^2(7K - 6)$ for L consecutive updates	Mixture of line, local, and global proposals
D3	Support loss along the line of action	$\sigma_{\parallel} \geq \ell_{\text{min}}$, end of each view	Fraction η along $\hat{\mathbf{u}}_k = \mathbf{R}_k^{\top} \hat{\mathbf{f}} / \ \hat{\mathbf{f}}\ $

where $\mathbf{B}(\mathbf{p}) = [\mathbf{I}_3; [\mathbf{p}]_{\times}]$. Marginalizing the follower variation by the Woodbury identity and the matrix determinant lemma gives the effective quantities $\tilde{\mathbf{T}}_k = \mathbf{T}_k - \mathbf{T}_k \mathbf{K}_k \mathbf{T}_k$, $\tilde{\mathbf{b}}_k = \mathbf{b}_k - \mathbf{T}_k \mathbf{K}_k \mathbf{b}_k$, $\mathbf{K}_k = (\sigma_f^{-2} \mathbf{I}_3 + \mathbf{T}_k)^{-1}$, and the profile likelihood

$$\hat{\mathbf{f}}(\mathbf{p}) = \left(\sum_k \mathbf{R}_k \tilde{\mathbf{T}}_k \mathbf{R}_k^{\top} \right)^{-1} \sum_k \mathbf{R}_k \tilde{\mathbf{b}}_k, \quad S(\mathbf{p}) = \sum_k \left(\tilde{c}_k - \tilde{\mathbf{b}}_k^{\top} \mathbf{R}_k^{\top} \hat{\mathbf{f}} \right) + \log \det \left(\mathbf{I}_3 + \sigma_f^2 \mathbf{T}_k \right), \quad (9)$$

with $\tilde{c}_k = \bar{\mathbf{r}}_k^{\top} \mathbf{W}_k \bar{\mathbf{r}}_k - \mathbf{b}_k^{\top} \mathbf{K}_k \mathbf{b}_k$. Sharing $\mathbf{f}$ across views is not needed for identifiability (Remark 1), but under (A2) it pools the force information of all views. The follower-variation model interpolates between the two cases: $\sigma_f \rightarrow 0$ recovers a strictly shared force, and $\sigma_f \rightarrow \infty$ independent per-view forces. Weights follow $w^{(i)} \propto \exp(-\frac{1}{2} S(\mathbf{p}^{(i)}))$, which reduces a six-dimensional search to a three-dimensional one. The normalizer $\log \det(\mathbf{I}_3 + \sigma_f^2 \mathbf{T}_k)$ grows with the moment arm and therefore adds a weak preference for short moment arms along the line of action; this preference is a property of the noise model, not information from the data, which is why the recovery of Section 3.5 is triggered by the data information rather than by the shape of the posterior.

The residual is averaged over 0.1-s sub-windows, and each sub-window triggers one update, i.e., 15 updates per 1.5-s measurement window. Because the particle set is regularized between updates, each update applies the incremental weight $\exp(-\frac{1}{2} [S_t(\mathbf{p}) - S_{t-1}(\mathbf{p})])$ evaluated at the current particle positions. Particles are roughened with a Gaussian kernel [27] of annealed width $\sigma_t = \max(\sigma_{\text{min}}, \sigma_0 0.9^t)$, $\sigma_0 = 5$ mm, $\sigma_{\text{min}} = 0.5$ mm, and resampled systematically when the effective sample size falls below $N/2$. We use $N = 3000$ in a search box $\mathcal{B} = [-0.15, 0.15] \times [-0.15, 0.15] \times [-0.05, 0.30]$ m in the tool frame. The posterior mean $\hat{\mathbf{p}}$, covariance $\Sigma_{\mathbf{p}}$, and the radius r_{95} of the ball around $\hat{\mathbf{p}}$ containing 95% of the weight summarize the measurement and its uncertainty.

3.5. Ambiguity-preserving recovery

Once a hypothesis disappears from a finite particle set, no later measurement can restore it. Three rules monitor different ways in which support is lost (Table 3).

D1: weight degeneracy When the effective sample size collapses, the particles are resampled systematically, and a fraction η of the copies is jittered around their parents with a kernel of $\max(3\sigma_t, 2 \text{ mm})$. D1 restores sample diversity around the current mode but cannot recover a truth that already lies outside the particle cloud.

D2: narrow but inconsistent A posterior can be narrow and have a healthy effective sample size while no particle explains the measurements; this is the path to a CBW measurement, and D1 does not detect it. D2 triggers when a narrow posterior ($r_{95} < r_{\text{nar}}$) remains inconsistent with the data, i.e., its best profile residual exceeds the 99% chi-square quantile with $7K - 6$ degrees of freedom, for L consecutive updates, so that transient outliers do not reset the filter. It re-supplies a fraction η of the particles in equal parts from three proposals: samples along the chord of $\mathcal{B}$ through $\hat{\mathbf{p}}$ in the direction of the current line of action (the geometrically meaningful escape direction), local perturbations of retained hypotheses with a 10-mm kernel, and samples drawn uniformly over $\mathcal{B}$ and resampled by their likelihood.

D3: line-of-action support preservation Between views, the posterior is correctly elongated along the unresolved direction (Proposition 1), and resampling thins it along exactly that direction. D3 is evaluated once at the end of each view and triggers on the information content of the data, not on the shape of the particle cloud: it computes the Cramér–Rao standard deviation $\sigma_{\parallel} = (\hat{\mathbf{u}}_k^{\top} \mathbf{I}_{p|f}^{-1} \hat{\mathbf{u}}_k)^{1/2}$ along the current line of action from the profile information of

Table 4

Frozen estimator parameters.

Parameter	Value	Parameter	Value
Particles N	3000	D2 persistence L	10 updates
D1 ratio ρ_{th}	0.02	D3 threshold $\ell_{\min}$	0.03 m
Re-supply fraction η	0.30	Min. roughening $\sigma_{\min}$	0.5 mm
D2 radius r_{nar}	0.05 m	Follower variation ε_f	3%
Observer gain K_O	50 s ⁻¹	Gate scale σ_g	0.1 N m

the views collected so far (Section 3.6), and fires when $\sigma_{\parallel} \geq \ell_{\min}$, i.e., when the data cannot yet localize the contact along the line. After a single view, $\sigma_{\parallel}$ is unbounded by Proposition 1, so D3 always fires, even if the particle cloud has already collapsed onto a short stretch of the line. It then replaces a fraction η of the particles by samples $\hat{\mathbf{p}} + s \hat{\mathbf{u}}_k + \mathbf{v}$, where s is uniform over the chord of $\mathcal{B}$ through $\hat{\mathbf{p}}$ along $\hat{\mathbf{u}}_k$ and $\mathbf{v}$ is a perturbation perpendicular to $\hat{\mathbf{u}}_k$ with the current perpendicular posterior spread. The direction $\hat{\mathbf{u}}_k = \mathbf{R}_k^T \hat{\mathbf{f}} / \|\hat{\mathbf{f}}\|$ is the theoretical line of action computed from the profiled force, not the empirical principal axis of the cloud, which is biased by the very thinning that D3 counteracts. All particles are then reweighted by the full accumulated likelihood. Because the injected particles lie in the null direction of the current view, the data of that view do not discriminate among them, apart from the weak normalizer preference of Section 3.4; only a new view can. D3 therefore does not resolve the ambiguity but keeps the correct hypothesis alive until the active view resolves it.

Execution order and guards Within each update, the filter roughens, profiles the force, updates the weights and diagnostics, evaluates D2 and then D1, and otherwise resamples when $\hat{N}_{\text{eff}} < N/2$; D3 is evaluated at the end of each view. A cooldown of five updates after each activation and caps of two activations per view and eight per episode prevent excessive recovery. Table 4 lists all parameters, which were frozen on a separate development set of 200 trials before the evaluation reported below.

3.6. Active measurement configuration

Profile Fisher information Linearizing the measurement model (3) in $\xi = (\mathbf{p}, \mathbf{f})$ gives the Jacobian

$$\mathbf{H}_k(\mathbf{p}, \mathbf{f}) = \begin{bmatrix} \mathbf{J}_k^T \begin{bmatrix} \mathbf{0}_3 \\ -[\mathbf{f}_k]_{\times} \end{bmatrix} & \mathbf{G}_k(\mathbf{p}) \end{bmatrix} \in \mathbb{R}^{7 \times 6}, \quad \mathbf{I}^{(\mathcal{K})} = \sum_{k \in \mathcal{K}} \mathbf{H}_k^T \tilde{\mathbf{W}}_k \mathbf{H}_k = \begin{bmatrix} \mathbf{I}_{pp} & \mathbf{I}_{pf} \\ \mathbf{I}_{fp} & \mathbf{I}_{ff} \end{bmatrix}, \quad (10)$$

where $\tilde{\mathbf{W}}_k$ is the view weight including the marginalized follower variation. Since the force is unknown but the quantity of interest is the contact point, we evaluate the profile (Schur-complement) information $\mathbf{I}_{p|f} = \mathbf{I}_{pp} - \mathbf{I}_{pf} \mathbf{I}_{ff}^{-1} \mathbf{I}_{fp}$, which discounts the part of the data that a change of force could explain [33].

Posterior-expected score Evaluating (10) at the posterior mean alone ignores that the posterior is a line whose points yield different information. For a candidate orientation $\mathbf{R}^c$, we therefore score the information expected over the posterior,

$$\mathcal{J}(\mathbf{R}^c) = \frac{1}{M} \sum_{i=1}^M \log \det \left(\mathbf{I}_{p|f}^{(1:k) \cup \{c\}} (\bar{\mathbf{p}}^{(i)}, \hat{\mathbf{f}}(\bar{\mathbf{p}}^{(i)})) + \varepsilon \mathbf{I}_3 \right) - \lambda_m \varphi(\mathbf{R}_k, \mathbf{R}^c), \quad (11)$$

where $\{\bar{\mathbf{p}}^{(i)}\}$ is a systematic subsample of size $M = 64$ of the posterior, $\varepsilon = 1 \text{ m}^{-2}$, φ is the rotation angle of the transition, and $\lambda_m = 1 \text{ rad}^{-1}$ penalizes large motions that would endanger the follower contact. The score evaluates the cumulative information of past views and the candidate jointly, so a candidate that repeats the current line of action is not rewarded.

Candidates inside the validated envelope The payload compensation is validated only up to the challenge tilt of Phase 1. Candidates are therefore generated relative to the initial orientation, not the current one: tilts of 10°, 20°, and 30° about 26 axes (the directions of a $3 \times 3 \times 3$ lattice). Candidates within 5° of an already visited view are dropped, and those that violate joint limits or would require more than 85% of a joint torque limit under the estimated payload and contact wrench are discarded.

Table 5Plant deviations from the nominal model (per trial). $\mathcal{N}$: normal, $\mathcal{U}$: uniform.

Error source	Distribution
Link masses	$\times(1 + \mathcal{N}(0, 0.03^2))$, clipped to $\pm 7.5\%$
Link centers of mass	$+\mathcal{N}(0, (3 \text{ mm})^2)$ per axis
Joint viscous damping	$\times(1 + \mathcal{U}(-0.2, 0.2))$
Residual dry friction	$\mathcal{U}(0.02, 0.06)$ / $\mathcal{U}(0.01, 0.03)$ N m (joints 1–4 / 5–7)
Torque constant	$\times(1 + \mathcal{N}(0, 0.01^2))$
Torque tracking noise	white, 0.2 / 0.05 N m (joints 1–4 / 5–7)
Encoder / velocity noise	2×10^{-5} rad / 2×10^{-3} rad s ⁻¹
Follower force	2% magnitude ripple, 1° direction wobble

Position-preserving reorientation The end effector moves to the selected orientation along eight inverse-kinematics waypoints on the rotation geodesic, interpolated by a cubic spline with a 2.0-s minimum-jerk time scaling, while its position is held at $\mathbf{o}_k$. Keeping the position fixed changes the object-frame force direction, and hence the moment-arm geometry, without dragging the object and its contact across the workspace, which preserves (A1) and (A3). The rotation can equally be performed about the current estimate $\hat{\mathbf{p}}$ instead of the tool center point, which shortens the path of the contact; Section 4.7 compares both with physical contact. Null-space motion alone is ineffective: if the end-effector pose is unchanged, so is the measurement geometry. Localization updates are suspended during the transition and resume after a settling time of 1.0 s, over a window of 1.5 s, consistent with (A4). An episode ends when at least two views have been taken and r_{95} falls below 10 mm without D3 firing, or after five views.

4. Experimental evaluation

4.1. Protocol

Simulation setup We simulated a Franka Research 3 in MuJoCo 3.13 at 1 kHz, with the link inertial parameters of the MuJoCo Menagerie FR3 model and a 0.73-kg Franka Hand, which constitute the nominal model used by the controller and the estimator. The simulated plant deviates from the nominal model in the ways that a commanded-torque-only arm does (Table 5). The arm is torque-controlled by a joint PD law with nominal gravity and Coriolis compensation; the observer runs on the nominal model with measured signals. Each trial draws a random initial configuration around the Franka ready pose, a rigid box-shaped object of $0.04\text{--}0.10 \times 0.04\text{--}0.10 \times 0.10\text{--}0.22$ m with a lateral offset of up to 30 mm and a mass of 0.3–1.5 kg, a contact point uniformly on its free surface, and a force of 5–25 N whose direction lies within 60° of the inward surface normal. Conditions that would require more than 80% of a joint torque limit at the initial orientation or at 30° tilts are redrawn. The contact force is applied at the material point and follows it during reorientation, with a 2% magnitude ripple and a 1° direction wobble (Ornstein–Uhlenbeck processes with a 0.5-s time constant) that emulate an imperfect follower. The Monte Carlo study comprises 1,200 paired trials per variant on test seeds that are disjoint from the 200 development trials used to freeze the design.

Metrics Position accuracy is reported as the median, mean, and 95th-percentile Euclidean error $e_p = \|\hat{\mathbf{p}} - \mathbf{p}\|$, and as the success rate $e_p < 10$ mm. Force accuracy is reported as the relative magnitude error and the angular error of $\hat{\mathbf{f}}$. The consistency of the reported uncertainty is assessed by the coverage of the 95% credible ellipsoid, i.e., the fraction of trials with $(\hat{\mathbf{p}} - \mathbf{p})^\top \Sigma_p^{-1} (\hat{\mathbf{p}} - \mathbf{p}) \leq \chi_{0.95}^2(3)$, a NEES-type test [35], and by the CBW rate.

Definition 1 (Confident-but-wrong measurement). An episode is CBW if its final posterior is confident, $r_{95} < r_{\text{conf}}$, while its estimate is wrong, $e_p > e_{\text{tol}}$. We use $r_{\text{conf}} = 10$ mm and $e_{\text{tol}} = 20$ mm.

Confidence and accuracy are thus evaluated separately: a CBW measurement is more harmful than a wide posterior, because downstream use has no indication that it should not be trusted. Efficiency is reported as the number of views, the time from contact onset to the end of localization, and the computation time of the next-view selection. Confidence intervals (95%) are obtained by bootstrap over trials.

Baselines and ablations All variants share the observer, the payload calibration, the uncertainty budget, and the frozen parameters of Table 4. (i) CPF without recovery; (ii) CPF with a single recovery rule D1, D2, or D3; (iii) D3

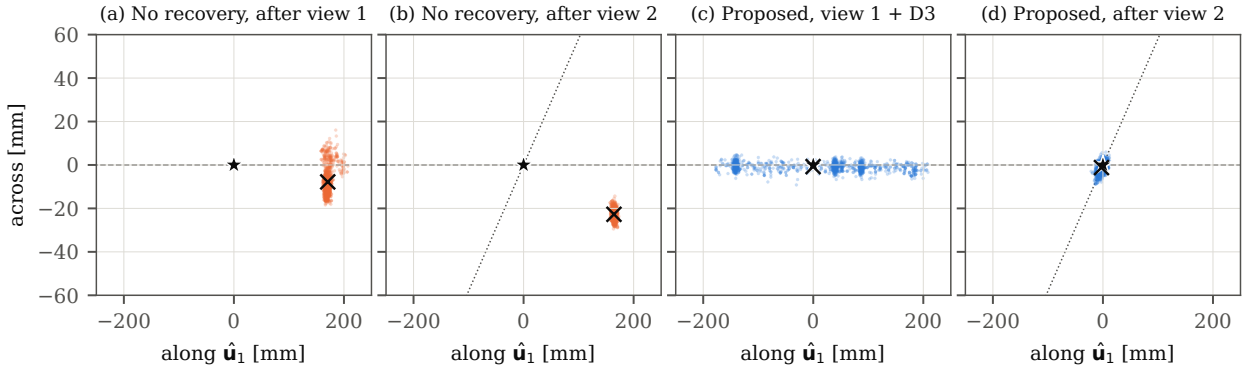


Figure 3: Representative paired episode (development seed), projected on the plane spanned by the first-view line of action (dashed) and the second-view line of action (dotted); star: true contact, cross: posterior mean. (a,b) Without recovery, the particles that survive the first view cover only a short stretch of the line, far from the truth; after the second view, the posterior is narrow and wrong (CBW, 166 mm error). (c,d) With recovery, D3 re-supplies hypotheses along the whole line after the first view, and the second view selects the true contact (3.2 mm error). The two axes are scaled differently.

re-supplying along the empirical principal axis $\mathbf{v}_1$ of Σ_p , triggered by cloud shape ($\lambda_1 \geq 3\lambda_2$ and $\sqrt{\lambda_1} \geq \ell_{\min}$), as a shape-based alternative to the information-based D3; (iv) next view chosen at random among feasible candidates, by maximal rotation, by FIM at the posterior mean, or by the posterior-expected score (11); (v) the proposed method (D1+D2+D3 with the posterior-expected score). Paired trials share the robot, object, contact point, force, model and actuation errors, and plant noise sequence, and every variant branches from the same post-calibration state.

4.2. Observability

Single view Single-view posteriors aligned with the theoretical line of action, as predicted by Proposition 1. The principal axis of Σ_p deviated from the true object-frame force direction by a median of 4.8° , with a median axis ratio $\sqrt{\lambda_1/\lambda_2}$ of 6. Perpendicular to the line, the single-view error had a median of 11.5 mm, whereas along the line it was bounded only by the extent of $\mathcal{B}$. Fig. 3 shows a representative episode.

Two views Table 6 and Fig. 2 report two views whose second orientation is rotated by ϑ about an axis perpendicular to the true force, with recovery enabled and no early stopping. The error is decomposed along the worst-information direction of Corollary 1, i.e., the bisector of the two lines of action, and in the plane orthogonal to it. In a noise-limited control study without model or actuation errors and with five times the torque noise, so that noise dominates the particle resolution, the RMS error along the worst direction followed the $1/\sin(\vartheta/2)$ scaling of (5) from 45° down to 10° (normalized values 1.47, 2.17, and 3.92 at 30° , 20° , and 10° , against 1.48, 2.20, and 4.39) and fell below it at 5° (6.63 against 8.78), where the error along the nearly parallel lines begins to be limited by the search region. Under the realistic errors of Table 5, the same component still grew steeply as the two lines approached each other, from 10.4 mm at 45° to 30.8 mm at 5° , whereas the orthogonal error stayed between 7.9 and 10.2 mm at every angle. This floor is the bias of the model and actuation errors, which the noise-based bound does not contain and which no choice of view can remove; it is why the total error saturates at moderate angles and why the candidate tilts need not exceed 30° . At $\vartheta = 0$, the second view repeats the first, and the error along the line is bounded only by the search region.

4.3. Recovery and measurement reliability

Table 7 isolates the recovery rules, and Fig. 5 shows the error distributions. Recovery reduced the p95 error from 137.6 mm to 21.9 mm (paired bootstrap 95% CI of the difference: 102.1–126.7 mm) and the CBW rate from 17.7% to 0.8% (10 of 1,200 trials; exact 95% CI 0.4–1.5%), while the median changed from 9.2 mm to 6.6 mm: recovery acts on the tail, where the measurement is most harmful. The coverage of the 95% credible ellipsoid rose from 68.5% to 81.9%. The reduction of the CBW rate does not hinge on the thresholds of Definition 1: with $(r_{\text{conf}}, e_{\text{tol}}) = (10, 10)$, (15, 30), and (20, 40) mm, the CBW rate was 23.0, 15.8, and 13.9% without recovery and 6.2, 0.4, and 0.2% with the proposed method.

Table 6

Two-view error versus the inter-view force angle θ (rotation axis perpendicular to $\mathbf{f}$). Worst direction: bisector of the two lines of action (Corollary 1). Normalized values are divided by their value at $\theta = 45^\circ$.

θ	0°	5°	10°	20°	30°	45°
Bound (5), normalized	∞	8.78	4.39	2.20	1.48	1.00
Noise-limited control study (300 trials)						
RMS error, worst direction, normalized	—	6.63	3.92	2.17	1.47	1.00
Realistic errors (300 trials)						
RMS error, worst direction [mm]	56.6	30.8	28.4	14.9	11.5	10.4
RMS error, orthogonal plane [mm]	7.9	9.0	10.2	8.9	9.4	9.9
Median total error [mm]	33.0	16.5	13.2	10.1	9.4	8.8

Table 7

Recovery ablation (simulation, posterior-expected next-view selection in all rows, 1,200 paired trials). Cov.: coverage of the 95% credible ellipsoid.

Recovery	Median [mm]	p95 [mm]	$e_p < 10$ mm [%]	CBW [%]	Cov. [%]	Views
None	9.2	137.6	53.8	17.7	68.5	4.27
D1	11.2	170.3	45.7	28.0	61.0	4.10
D2	7.2	22.3	68.6	1.2	77.7	4.22
D3, shape-based (ν_1)	7.2	132.6	64.8	12.2	79.6	4.41
D3, information-based ($\hat{\mu}_k$)	7.0	31.9	67.8	6.6	81.2	4.42
D1+D2+D3 (proposed)	6.6	21.9	71.7	0.8	81.9	4.21

The classical effective-sample-size rule D1 alone made the tail worse (p95 170.3 mm, CBW 28.0%). The dominant failures are therefore not a collapse of the weights, but the loss of support along the unresolved line of action and narrow posteriors that no longer explain the data. The force dependence of the failures confirms this mechanism. Without recovery, the CBW rate grew with the force magnitude, from 0.7% at 5–10 N to 40.7% at 20–25 N (38.8% in the upper third of force magnitudes): the larger the force, the sharper the likelihood perpendicular to the line, and the fewer particles survive on it. With recovery, the CBW rate stayed at or below 2.5% in every force bin (1.8% in the upper third).

The two targeted rules address the two failures on different time scales. D2 is a safety net that acts once a narrow posterior has persistently contradicted the data; alone, it reduced the final p95 error to 22.3 mm with a CBW rate of 1.2%. D3 acts at the end of the first view, before any wrong commitment, and makes the second view count for the bulk of the trials: after two views, 61.2% of the trials were below 10 mm with D3 alone, against 44.8% with D2 alone and 63.1% with both. The tail after two views is set by the trials that D2 repairs later, with p95 errors of 35.5 mm for D3 alone, 33.0 mm for D2 alone, and 25.9 mm for both. Combining the rules gave the lowest final CBW rate and p95 error of all variants. How D3 is triggered and where it re-supplies matter. The shape-based variant, which re-supplies along the empirical principal axis only while the cloud is still elongated, barely improved the tail (p95 132.6 mm, CBW 12.2%): by the end of the first view, the surviving cloud has often already collapsed onto a short stretch of the line, is no longer elongated, and the shape test does not fire.

Fig. 4 extends the coverage test to all credible levels. With recovery, the empirical coverage follows the nominal level closely up to intermediate levels and is moderately optimistic in the far tail (81.9% at the nominal 95%); without recovery, the reported uncertainty is optimistic at every level, because a lost hypothesis cannot be represented by any spread of the surviving particles.

4.4. Active sensing

Table 8 crosses recovery with the next-view strategy, and Fig. 6 shows the error after each view. Without recovery, neither strategy removes the tail, because the true hypothesis is often lost before the second view is taken. With recovery, both strategies eventually reach similar final errors under the stopping rule, but the active strategy stops earlier (4.21 versus 4.64 views on average; 17.0 s versus 18.9 s from contact onset). At a fixed budget, the difference is large: after two views, the p95 error was 25.9 mm with the active view and 40.4 mm with a random view (paired difference 14.4 mm, 95% CI 10.9–20.5 mm), and 63.1% versus 45.6% of the trials were below 10 mm. Recovery preserves the

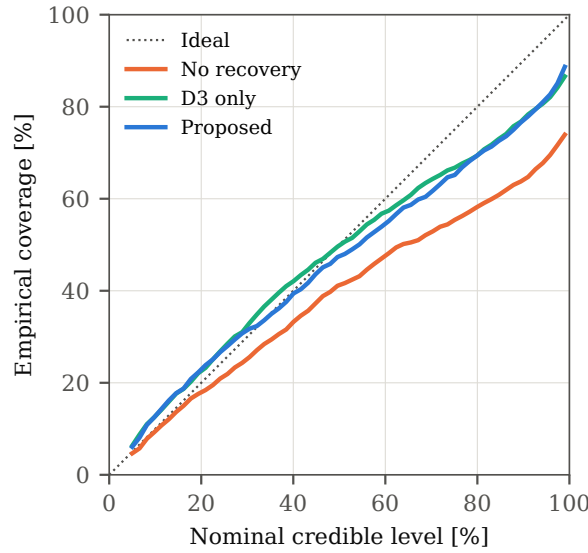


Figure 4: Reliability of the reported uncertainty: empirical coverage of the credible ellipsoid versus its nominal level (NEES test, 1,200 trials).

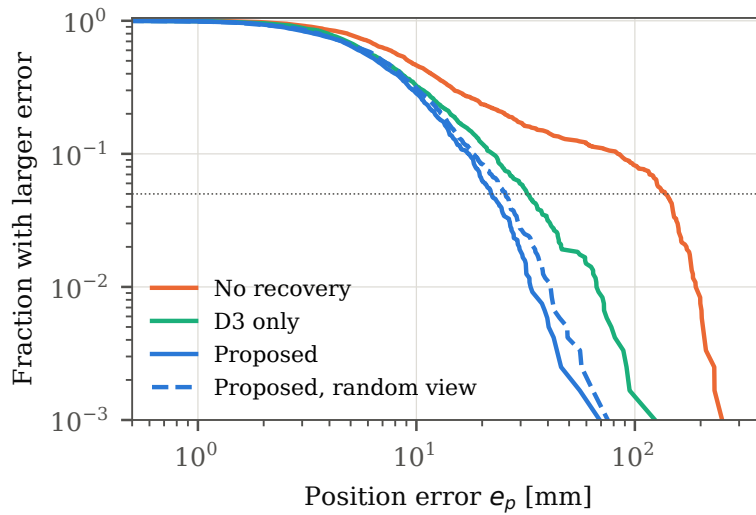


Figure 5: Complementary cumulative distribution of the position error over 1,200 paired trials (log-log). The dotted line marks the 95th percentile.

right hypothesis; active sensing separates it with the fewest measurements. The stopping threshold trades accuracy for speed: with $r_{95} < 20$ mm instead of 10 mm, localization took 2.83 views and 10.7 s instead of 4.21 views and 17.0 s, at a median error of 7.3 mm, a p95 error of 22.3 mm, and a CBW rate of 2.8% for the correspondingly relaxed confidence radius of 20 mm.

Among the view-selection rules (Table 9), maximal rotation performed like random selection: large rotations about axes nearly parallel to the force add motion without adding information (Proposition 2). The posterior-expected score performed on par with the FIM evaluated at the posterior mean (after two views, p95 25.9 mm versus 25.2 mm), so the cheaper mean-point score is an equally valid choice in this regime. On a single idle CPU core, generating and checking the 78 feasible candidates by inverse kinematics took 303 ms, and scoring them took 99 ms with the posterior-expected score and 10 ms with the mean-point score. Because the candidate set is fixed relative to the initial orientation, the

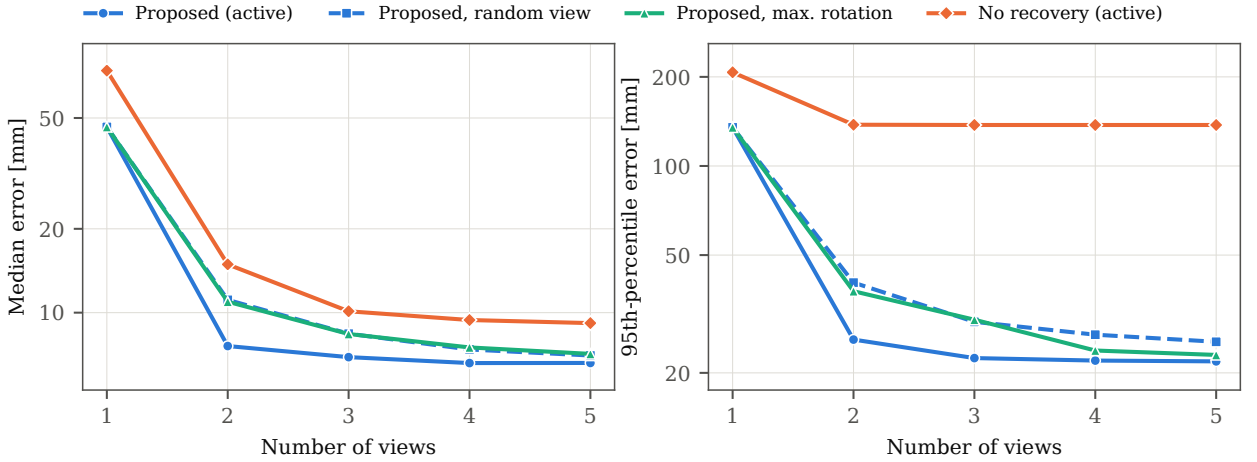
Table 8Recovery $\times$ next-view strategy (simulation, 1,200 paired trials per cell).

Recovery	Next view	p95 [mm]	CBW [%]	$e_p < 10$ mm [%]	Views
Off	Random	154.8	17.5	44.7	4.51
Off	Active	137.6	17.7	53.8	4.27
On	Random	25.5	1.7	68.9	4.64
On	Active	21.9	0.8	71.7	4.21

Table 9

Next-view selection rules (simulation, recovery on, 1,200 paired trials). Selection: mean wall time per selection, including the inverse kinematics of all candidates, measured during the parallel Monte Carlo runs; single-core times are given in the text.

Next view	Median [mm]	p95 [mm]	CBW [%]	Views	Selection [s]
Random feasible	7.0	25.5	1.7	4.64	1.12
Maximal rotation	7.1	23.0	1.6	4.49	1.12
FIM at posterior mean	6.6	22.8	1.1	4.20	1.18
Posterior-expected FIM (11)	6.6	21.9	0.8	4.21	1.58

**Figure 6:** Median and 95th-percentile position error after each view (1,200 paired trials; trials that stopped earlier keep their final estimate). With recovery, the actively selected second view already achieves the accuracy that random or maximal rotations reach only after four to five views.

inverse kinematics can be computed once per episode, which leaves the scoring well inside the 1-s settling period that follows each reorientation.

4.5. Robustness, sensitivity, and computation

The accuracy scaled with the force magnitude as predicted by the $1/\|f\|$ factor of (5) (Table 10): the median error fell from 9.8 mm at 5–10 N to 5.0 mm at 20–25 N. The force itself was measured with a median magnitude error of 1.3% and a median angular error of 1.0° . Heavy payloads did not degrade the measurement, and neither did a threefold residual friction (median 6.5 mm, CBW 0.0%). A link-mass error of 10%, which the four-parameter payload calibration can absorb only locally, raised the median error to 9.7 mm and the CBW rate to 2.3%. When the true errors exceeded the values assumed in the uncertainty budget, the accuracy degraded gracefully but the reported uncertainty became optimistic. A torque-constant error of 3% (budget 1%) raised the median error to 11.4 mm and lowered the coverage from 81.9% to 57.7%, and a follower variation of 5% and 3° (budget 3%) lowered the coverage to 68.3% at a median error of 7.1 mm. The CBW rate stayed at or below 1.7% in both cases, because recovery does not rely on an exact budget.

Table 10

Stress conditions (simulation, proposed method). Force errors are medians. Force bins and the heavy-payload row are subsets of the main study; the other rows are separate studies. In the last two rows, the true error exceeds the value assumed by the uncertainty budget.

Condition	Trials	Median [mm]	p95 [mm]	CBW [%]	Cov. [%]	$ \Delta\ f\ $ [%]	$\angle\Delta f$ [deg]
$\ f\ \in [5, 10)$ N	288	9.8	30.3	0.3	85.4	2.0	1.6
$\ f\ \in [10, 15)$ N	324	7.0	18.5	0.0	86.1	1.4	1.1
$\ f\ \in [15, 20)$ N	283	5.7	18.1	2.5	79.5	1.1	0.9
$\ f\ \in [20, 25]$ N	305	5.0	13.9	0.7	76.4	1.0	0.8
Payload ≥ 1.1 kg	418	6.5	23.7	0.5	82.8	1.5	1.1
Link-mass error 10%	300	9.7	28.4	2.3	80.0	2.0	1.7
Residual friction $\times 3$	300	6.5	21.0	0.0	82.3	1.2	0.9
Torque constant 3% (budget 1%)	300	11.4	36.9	1.7	57.7	2.4	1.7
Follower 5%, 3° (budget 3%)	300	7.1	23.9	1.0	68.3	1.9	1.8

Recovery is insensitive to most of its frozen parameters. In a separate study of 300 trials that changed one parameter at a time, the re-supply fraction ($\eta = 0.1$ and 0.5), the D3 threshold ($\ell_{\min} = 1$ and 10 cm), and a shorter D2 persistence ($L = 5$) kept the p95 error between 19.5 and 22.8 mm and the CBW rate at or below 1.0%. A longer persistence ($L = 20$) raised the CBW rate to 6.3%: it exceeds the 15 updates of one measurement window, so that D2 can no longer act within the view in which an inconsistency arises. The persistence must therefore stay below the number of updates per window.

With 500, 1000, 3000, and 5000 particles (400 matched trials), the p95 error was 22.1, 19.9, 19.2, and 18.9 mm and the CBW rate 2.2, 1.8, 0.2, and 1.0%. One filter update took 4.1, 7.7, 22.4, and 37.9 ms on a single CPU core in our Python implementation, so that $N = 3000$ uses less than a quarter of the 100-ms update period.

4.6. Trust gate

The gate study comprised 1,000 trials whose model errors were drawn from a mixture of link-mass errors of 2–15%, center-of-mass errors of 2–10 mm, torque-constant errors of 0–8%, and unmodeled joint springs of 0–3 N m/rad. A trial was labeled invalid when the RMS static compensation error over 40 random orientations inside the validated envelope exceeded 0.25 N m, which applied to 90.4% of the trials. One half of the trials derived the threshold by the lexicographic rule of Section 3.3. On the other half, the frozen gate ($\gamma = 0.87$) admitted 0 of the 456 invalid models (exact 95% upper bound 0.8%) and 0 CBW localizations, and it accepted 52.3% of the 44 valid models; the statistic separated valid from invalid models with an AUC of 0.99. The gate is deliberately conservative: rejecting a valid model delays the measurement, whereas admitting an invalid one produces a wrong measurement without warning. The gate screens exactly the trials that would yield the least reliable measurements: over the whole study, the median localization error was 7.1 mm for valid and 17.3 mm for invalid models, with CBW rates of 0.0% and 2.1%.

4.7. Physical contact and pusher compliance

The studies above apply the contact force as an ideal follower force at the material point. To test this abstraction, 300 additional trials replaced it by a physical contact: a pusher with a 6-mm spherical tip and a friction coefficient of 1.0 pressed on the object surface through the contact model of MuJoCo, and the contact force followed from the collision. The pusher applied the commanded world force either as a pure force controller that follows the surface (follower) or with a lateral impedance of 50, 150, or 300 N/m around its initial position (anchored), which models a pusher that does not follow the moving surface. The initial force direction was drawn within 15° of the inward normal, and the object was rotated about the tool center point as before, about the current estimate $\hat{p}$, or, as an ideal reference, about the true contact point (Table 11).

With a following pusher, the physical contact reproduced the results of the ideal force model: the median error was 7.8 mm, the p95 error 20.4 mm, and the CBW rate 0.7%. The force changed by a median of 3.6% between windows, and the tip slid by a median of 3.1 mm, because the constant world force approaches the edge of the friction cone as the surface rotates; in 0.7% of the trials, the accumulated rotation took the force outside the cone, and the tip slid off the face. Rotating about the estimate instead of the tool center point shortened the distance that the pusher had to follow from 48 to 24 mm at unchanged accuracy.

Table 11

Physical contact (simulation, proposed estimator, 300 paired trials per row). Pivot: point about which the object is rotated. Travel: median of the largest displacement of the pusher tip during a reorientation. Force change: median of the largest change of the window-mean contact force relative to the first window.

Pusher	Pivot	Median [mm]	p95 [mm]	CBW [%]	Contact lost [%]	Travel [mm]	Force change [%]
Follower	Tool center	7.8	20.4	0.7	0.7	48	3.6
Follower	Estimate	7.8	18.9	0.7	0.7	24	3.6
Follower	True contact	7.6	19.3	0.7	0.7	2	3.6
50 N/m	Tool center	9.7	32.2	8.3	0.3	47	9.7
150 N/m	Tool center	13.7	175.9	19.7	1.0	46	26.7
300 N/m	Tool center	22.2	190.1	38.0	2.7	44	48.5
300 N/m	Estimate	24.6	301.7	40.3	27.3	29	44.1
300 N/m	True contact	7.7	20.9	1.7	0.0	2	4.4

[Placeholder — hardware]

FR3 with grasped object, force-controlled pusher with F/T reference;
first-view posterior, active rotation, final posterior.

Figure 7: Real-robot setup and a representative localization episode.

A pusher that does not follow turns the motion of the material point into a lateral force. The force then differed between views by a median of 9.7, 26.7, and 48.5% for lateral stiffnesses of 50, 150, and 300 N/m, which violates assumption (A2) and raised the CBW rate to 8.3, 19.7, and 38.0%. Rotating about the true contact point removed the problem (CBW 1.7%), but rotating about the estimate did not: after one view, the estimate is uncertain along the line of action by construction (Proposition 1), so an informative rotation about it still moves the true contact, and with a wrong pivot, the pusher lost contact in 27.3% of the trials. Contact-preserving measurement therefore requires a pusher that is compliant tangentially to the surface, such as a human hand or a force-controlled pusher without lateral position feedback; this is the practical meaning of assumption (A3).

4.8. Real-robot validation

Setup and ground truth The estimator ran on a Franka Research 3 using only $\mathbf{q}$, $\dot{\mathbf{q}}$, and the commanded torque of the torque-control interface, reconstructed as the user command plus the gravity torque that the controller adds internally; the arm's joint-torque sensors were logged for diagnostics but never used by the estimator. Because the FR3 closes its joint torque loop internally, its actuation is more faithful than that of a current-controlled drive; to emulate the latter, 24 additional trials scaled the commanded torques in software by random per-joint factors of up to $\pm 3\%$ that were unknown to the estimator. A 3D-printed object, whose geometry was withheld from the estimator, was held in the gripper. The contact was applied by a second manipulator carrying a spherical tip on an ATI Mini45 F/T sensor under force control, which followed the same material point during reorientation (A3). The reference contact point was obtained from the pusher kinematics, registered to the object by touching fiducial points, and the reference force from the pusher F/T sensor transformed to the world frame. Neither reference entered the estimator.

Protocol We used eight contact points on the object surface, including points near edges and near the center of mass, and three force settings of 8, 15, and 22 N in lateral and diagonal directions. Each of the 24 conditions was repeated

Table 12Real-robot results. **Franka Research 3, 8 contact points, 3 force settings.**

Variant	Trials	Median [mm]	p95 [mm]	CBW [%]	$ \Delta\ f\ $ [%]	$\angle\Delta f$ [deg]
No recovery (replay)	96	12.8	131.2	14.6	7.2	5.6
Recovery, random view (live)	48	9.9	35.7	2.1	6.4	5.1
Proposed (live)	96	8.7	26.3	1.0	6.1	4.9

four times with the proposed method (96 trials), and all measurement windows were logged. The recovery ablation was evaluated by replaying the logged windows of the same 96 trials, which makes it paired with the proposed method on identical data. Random next-view selection changes the physical motion and was therefore run live on 48 trials covering all 24 conditions twice. The Phase-1 protocol and all estimator parameters were identical to the simulation study (Table 4).

Results Table 12 summarizes the hardware results. The proposed method attained a median position error of 8.7 mm and a p95 error of 26.3 mm, with 58.3% of trials below 10 mm. The force was measured with a median magnitude error of 6.1% and a median angular error of 4.9°. One of 96 trials was CBW (1.0%), and the 95% credible ellipsoid contained the reference point in 78.1% of trials. Single-view posteriors aligned with the theoretical line of action within a median of 6.3°. Localization required 4.4 views and 19.6 s on average. The ordering of the variants matched simulation: replaying the same windows without recovery raised the CBW rate to 14.6%, and random views raised the p95 error to 35.7 mm. Across the four repetitions of each condition, the standard deviation of the estimate was 3.4 mm (median over conditions), which characterizes the repeatability of the sensorless measurement.

5. Discussion

What is and is not identifiable The analysis separates two sources of error that are usually conflated. Noise and model error limit the accuracy perpendicular to the line of action and scale as $\sigma_m/\|f\|$; the component along the line is not limited by noise at all but is structurally unobservable from one view. Improving sensors, observers, or filters cannot remove the second component; changing the measurement configuration can, and (5) states by how much. The force bins of Table 10 follow the first component, and the angle sweep of Table 6 the second.

Why effective-sample-size recovery fails The effective sample size measures how evenly the weights are spread, not whether the particles span the directions that future measurements will resolve. After the first view, the weight concentrates on the few particles that happen to lie near the line, so the classical rule fires; resampling then copies these particles, and the posterior becomes both shorter along the line and more confident. This is why D1 alone increased the CBW rate. The targeted rules instead ask what the data can and cannot resolve: D3 restores the direction that the data cannot resolve yet, and D2 detects a narrow posterior that contradicts the data. The two failures occur in different trials, which is why each rule alone leaves a higher CBW rate than their combination.

Measurement reliability The CBW rate and the credible-region coverage capture what matters for downstream use of a sensorless measurement: whether the reported uncertainty can be trusted [3]. The uncertainty budget is what makes the particle posterior usable as a measurement result. Its Type B terms describe effects that contact-free calibration cannot see, and when the true torque-constant error or follower variation exceeds the values assumed in the budget, the coverage drops (Table 10). On a current-controlled arm, u_k should therefore be taken from the tolerance of the drive's torque constant, and ε_f from the specification of the pusher or the expected variability of a human hand.

Limitations The method assumes a single follower contact at a fixed material point with an approximately constant world-frame force during an episode. Sliding contacts, contact switching, multiple simultaneous contacts, and contacts with a fixed environment, whose force direction changes with the reorientation, violate these assumptions and are outside the present scope. The pusher must be compliant tangentially to the surface, because a laterally stiff pusher turns the motion of the contact into force changes between views. Low forces remain the hardest regime, as predicted by the $1/\|f\|$ scaling, and the motion is confined to the envelope validated in Phase 1. The information criterion is local; strongly multimodal posteriors may benefit from a non-myopic objective.

6. Conclusion

We presented a sensorless measurement of contact location and force on an unknown grasped object that uses only joint states and commanded torques. We showed that the single-view measurement is structurally ambiguous along the line of action, that two transverse views make it identifiable, and how the attainable accuracy scales with force magnitude and view separation. The estimator combines a validated residual with a GUM-type uncertainty budget and a force-profiled particle filter that preserves the physically valid ambiguity, detects when its support becomes unreliable, and actively changes the measurement configuration until the contact is observable. In 1,200 paired simulation trials with realistic model and actuation errors, recovery reduced the 95th-percentile error by a factor of 6.3 and the confident-but-wrong rate from 17.7% to 0.8%, and the actively selected second view reached a 95th-percentile error of 25.9 mm, against 40.4 mm for a random second view. **On the real robot, the method localized contacts with a median error of 8.7 mm against an independent reference.**

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The simulation code, the per-trial results that support the findings, and supplementary material are available on the project page, <https://anonymous.4open.science/w/apcl/>.

References

- [1] S. Haddadin, A. De Luca, A. Albu-Schäffer, Robot collisions: A survey on detection, isolation, and identification, *IEEE Transactions on Robotics* 33 (6) (2017) 1292–1312. doi:10.1109/TR0.2017.2723903.
- [2] A. Bicchi, J. K. Salisbury, D. L. Brock, Contact sensing from force measurements, *The International Journal of Robotics Research* 12 (3) (1993) 249–262. doi:10.1177/027836499301200304.
- [3] JCGM, Evaluation of measurement data – Guide to the expression of uncertainty in measurement, Tech. Rep. JCGM 100:2008, Joint Committee for Guides in Metrology (BIPM), Sèvres, France (2008). doi:10.59161/JCGM100-2008E.
- [4] A. De Luca, R. Mattone, Sensorless robot collision detection and hybrid force/motion control, in: *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2005, pp. 999–1004. doi:10.1109/ROBOT.2005.1570247.
- [5] A. De Luca, A. Albu-Schäffer, S. Haddadin, G. Hirzinger, Collision detection and safe reaction with the DLR-III lightweight manipulator arm, in: *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2006, pp. 1623–1630. doi:10.1109/IROS.2006.282053.
- [6] B. Eren, Event-level collision detection for industrial robotic systems using deep temporal learning on internal sensor signals, *Measurement* 289 (2026) 122712. doi:10.1016/j.measurement.2026.122712.
- [7] L. Manuelli, R. Tedrake, Localizing external contact using proprioceptive sensors: The contact particle filter, in: *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2016, pp. 5062–5069. doi:10.1109/IROS.2016.7759743.
- [8] D. Zurlo, T. Heitmann, M. Morlock, A. De Luca, Collision detection and contact point estimation using virtual joint torque sensing applied to a cobot, in: *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2023, pp. 7533–7539. doi:10.1109/ICRA48891.2023.10160661.
- [9] J. K. Salisbury, Interpretation of contact geometries from force measurements, in: *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, Vol. 1, 1984, pp. 240–247. doi:10.1109/ROBOT.1984.1087180.
- [10] A. Wahrburg, J. Bös, K. D. Listmann, F. Dai, B. Matthias, H. Ding, Motor-current-based estimation of Cartesian contact forces and torques for robotic manipulators and its application to force control, *IEEE Transactions on Automation Science and Engineering* 15 (2) (2018) 879–886. doi:10.1109/TASE.2017.2691136.
- [11] E. Magrini, F. Flacco, A. De Luca, Estimation of contact forces using a virtual force sensor, in: *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2014, pp. 2126–2133. doi:10.1109/IROS.2014.6942848.
- [12] S. Liu, L. Wang, X. V. Wang, Sensorless force estimation for industrial robots using disturbance observer and neural learning of friction approximation, *Robotics and Computer-Integrated Manufacturing* 71 (2021) 102168. doi:10.1016/j.rcim.2021.102168.
- [13] R. Ding, X. Mu, M. Cheng, B. Xu, G. Li, Terminal force soft sensing of hydraulic manipulator based on the parameter identification, *Measurement* 200 (2022) 111551. doi:10.1016/j.measurement.2022.111551.
- [14] T. N. Truong, S. Kim, High-performance fixed-time active fault-tolerant control for robotic manipulators via sparse physics-informed dynamics learning, *Information Sciences* 758 (2027) 124001. doi:10.1016/j.ins.2026.124001.
- [15] J. Xu, X. Li, Z. Yang, J. Su, R. Chen, D. Shang, Transmission friction measurement and suppression of dual-inertia system based on RBF neural network and nonlinear disturbance observer, *Measurement* 202 (2022) 111793. doi:10.1016/j.measurement.2022.111793.
- [16] N. Likar, L. Žlajpah, External joint torque-based estimation of contact information, *International Journal of Advanced Robotic Systems* 11 (7) (2014) 107. doi:10.5772/58834.

- [17] M. Iskandar, O. Eiberger, A. Albu-Schäffer, A. De Luca, A. Dietrich, Collision detection, identification, and localization on the DLR SARA robot with sensing redundancy, in: Proc. IEEE Int. Conf. Robotics and Automation (ICRA), 2021, pp. 3111–3117. doi:10.1109/ICRA48506.2021.9561677.
- [18] A. Zwiener, C. Geckeler, A. Zell, Contact point localization for articulated manipulators with proprioceptive sensors and machine learning, in: Proc. IEEE Int. Conf. Robotics and Automation (ICRA), 2018, pp. 323–329. doi:10.1109/ICRA.2018.8462869.
- [19] J. Liang, O. Kroemer, Contact localization for robot arms in motion without torque sensing, in: Proc. IEEE Int. Conf. Robotics and Automation (ICRA), 2021, pp. 6322–6328. doi:10.1109/ICRA48506.2021.9562058.
- [20] T. Pang, J. Umenberger, R. Tedrake, Identifying external contacts from joint torque measurements on serial robotic arms and its limitations, in: Proc. IEEE Int. Conf. Robotics and Automation (ICRA), 2021, pp. 6476–6482. doi:10.1109/ICRA48506.2021.9561761.
- [21] C. G. Atkeson, C. H. An, J. M. Hollerbach, Estimation of inertial parameters of manipulator loads and links, The International Journal of Robotics Research 5 (3) (1986) 101–119. doi:10.1177/027836498600500306.
- [22] H. Kim, S. Kim, J. Park, Enhanced online parameter estimation of unknown objects via sparse identification of unmodeled dynamics, Journal of Mechanical Science and Technology 39 (5) (2025) 2343–2354. doi:10.1007/s12206-025-2201-2.
- [23] C. Wang, X. Zang, X. Zhang, Y. Liu, J. Zhao, Parameter estimation and object gripping based on fingertip force/torque sensors, Measurement 179 (2021) 109479. doi:10.1016/j.measurement.2021.109479.
- [24] C. Gaz, M. Cognetti, A. Oliva, P. Robuffo Giordano, A. De Luca, Dynamic identification of the Franka Emika Panda robot with retrieval of feasible parameters using penalty-based optimization, IEEE Robotics and Automation Letters 4 (4) (2019) 4147–4154. doi:10.1109/LRA.2019.2931248.
- [25] J. Jia, X. Yan, Y. Liao, K. An, D. Jiang, Physically consistent inertial parameter identification of AUVs via geometric methods, Measurement 275 (2026) 121365. doi:10.1016/j.measurement.2026.121365.
- [26] M. S. Arulampalam, S. Maskell, N. Gordon, T. Clapp, A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking, IEEE Transactions on Signal Processing 50 (2) (2002) 174–188. doi:10.1109/78.978374.
- [27] C. Musso, N. Oudjane, F. Le Gland, Improving regularised particle filters, in: A. Doucet, N. de Freitas, N. Gordon (Eds.), Sequential Monte Carlo Methods in Practice, Springer, New York, 2001, pp. 247–271. doi:10.1007/978-1-4757-3437-9_12.
- [28] F. Pukelsheim, Optimal Design of Experiments, Vol. 50 of Classics in Applied Mathematics, Society for Industrial and Applied Mathematics, Philadelphia, PA, 2006, unabridged republication of the 1993 Wiley edition. doi:10.1137/1.9780898719109.
- [29] Y. Yin, D. Gao, Y. Lu, C. Zhao, G. Li, K. Deng, In-situ robot joint stiffness identification using an eye-in-hand camera with optimal measurement pose selection, Measurement 253 (2025) 117579. doi:10.1016/j.measurement.2025.117579.
- [30] J. Bohg, K. Hausman, B. Sankaran, O. Brock, D. Kragic, S. Schaal, G. S. Sukhatme, Interactive perception: Leveraging action in perception and perception in action, IEEE Transactions on Robotics 33 (6) (2017) 1273–1291. doi:10.1109/TR0.2017.2721939.
- [31] A. Petrovskaya, O. Khatib, Global localization of objects via touch, IEEE Transactions on Robotics 27 (3) (2011) 569–585. doi:10.1109/TR0.2011.2138450.
- [32] S. Zhong, N. Fazeli, D. Berenson, RUMI: Rummaging using mutual information, IEEE Transactions on Robotics 41 (2025) 5431–5450. doi:10.1109/TR0.2025.3605251.
- [33] A. Fischer, Fisher information and Cramér-Rao bound for unknown systematic errors, Measurement 113 (2018) 131–136. doi:10.1016/j.measurement.2017.08.042.
- [34] E. Garcia, T. Hausotte, A. Amthor, Bayes filter for dynamic coordinate measurements – Accuracy improvement, data fusion and measurement uncertainty evaluation, Measurement 46 (9) (2013) 3737–3744. doi:10.1016/j.measurement.2013.04.001.
- [35] Y. Bar-Shalom, X.-R. Li, T. Kirubarajan, Estimation with Applications to Tracking and Navigation: Theory, Algorithms and Software, Wiley, New York, 2001. doi:10.1002/0471221279.